MILITARY STANDARD

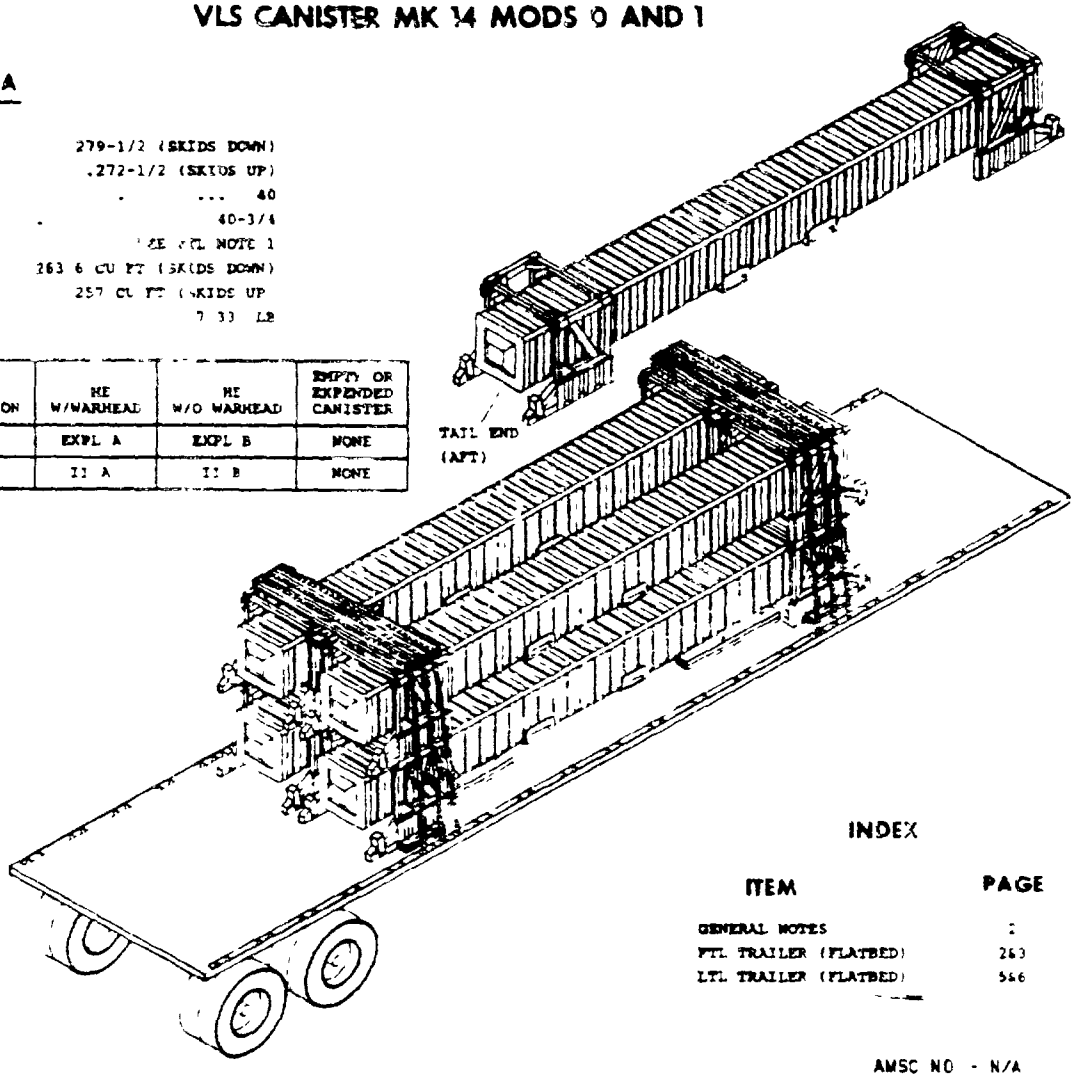
MIL-STD-1320-223B
(NAVY)
11 OCTOBER 1989
SUPERSEDING
MIL-STD-1320-223A (NAVY)
20 DECEMBER 1985

TRUCKLOADING
GUIDED MISSILE RGM-109
(TOMAHAWK)
VLS CANISTER MK 14 MODS 0 AND 1

CONTAINER DATA

DIMENSION	
LENGTH	279-1/2 (SKIDS DOWN)
LENGTH	272-1/2 (SKIDS UP)
WIDTH	40
HEIGHT	40-3/4
WEIGHT	SEE FTL NOTE 1
CUBE	263.6 CU FT (SKIDS DOWN)
CUBE	257 CU FT (SKIDS UP)
WEIGHT	7 33 LB

HAZARD CLASSIFICATION	MI W/WARHEAD	MI W/O WARHEAD	EMPTY OR EXPENDED CANISTER
DOT	EXPL A	EXPL B	NONE
CG	II A	II B	NONE



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL TRAILER (FLATBED)	263
LTL TRAILER (FLATBED)	566

AMSC NO - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE
DISTRIBUTION IS UNLIMITED

REV LTR	REVISION DESCRIPTION	DATE	TDA APPROVAL	SYSCOM APPROVAL
B	REDESIGN SEPARATORS			
A	ADDED TARPAULIN REQUIREMENT AND REVISED CANISTER ORIENTATION	12/20/85	5/MSG	5/JRK

NOTES
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED
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SIGNATURE TDA APPSTA EARLE NJ DATE 10/11/89

SIGNATURE SCS SPT BY DIRECTION DATE 10/11/89

ORIGINATOR [signature] 10/11/89 DATE

APPSTA EARLE, NEW JERSEY

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GENERAL NOTES

1 THIS DRAWING PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING VLS CANISTERS LOADED WITH THE TOMAHAWK MISSILE THE VLS CANISTERS SHALL BE EQUIPPED WITH ITS STACKING FRAME HARDWARE THE SKID ENDS OF THE VLS CANISTER SHALL BE IN THE "UP" POSITION

2 THE PROCEDURES AND PRACTICES CONTAINED HEREIN ARE INTENDED FOR 35-FEET AND LONGER TANDEM AXLE FLATBED TRAILERS DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS TRAILER AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER)

3 HANDLE ALL CANISTERS ONE AT A TIME DO NOT HANDLE TWO HIGH WHEN THE CANISTERS ARE STACKED TWO HIGH THE TOP CANISTER MAY BE BOLTED TO THE BOTTOM CANISTER BY BOLTING THE BOTTOM CANISTER STACKING FRAMES TO THE TOP CANISTER SKIDS USING THE BOLTS PROVIDED (FOUR BOLTS EACH END) BOLTING IS NOT MANDATORY

4 THE FWD ENDS OF THE CONTAINERS SHOULD BE AT THE FWD END OF THE TRAILER

5 DIMENSION X FOR 35 FT 108 5
DIMENSION X FOR 40 FT TRAILER 138 5
DIMENSION X FOR 42 FT TRAILER ... .150 5
DIMENSION X FOR 44 FT TRAILER . 162 5
DIMENSION X FOR 45 FT TRAILER . . .168 5
DIMENSION X FOR 48 FT TRAILER . 186 5

6 A STACK OF FOUR CANISTERS REQUIRES SIX TIEDOWNS TWO CANISTERS REQUIRE FOUR TIEDOWNS ONE CANISTER REQUIRES TWO TIEDOWNS (SEE PAGES 3 5 AND 6)

7 CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASIS FOR TIEDOWNS

8 STEEL STRAPPING USED AS TIEDOWNS SHALL BE 2 X 050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

9 CHAINS, FITTING AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL)

10 PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIEDOWNS WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2-INCH STRAPPING STAPLES WHEN USING CHAIN THE CHAINS SHALL BE SECURED TO THE PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626 THE CHAINS FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRENGTH GOUGING BENT LINKS WEAR AND ANY OTHER NOTICEABLE DEFECTS THE INSPECTOR SHALL CONFIRM THAT THE CHAINS FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO 22 OF DD FORM 626 ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER

11 UNLESS OTHERWISE SPECIFIED DUNNAGE LUMBER STEEL STRAPPING STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY)

12 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

13 AFTER THE BLOCKING BRACING AND TIEDOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT WATERPROOF TARPAULINS THE TARPAULINS MAY BE UNDER THE TIEDOWNS

14 AFTER THE TARPAULINS ARE IN PLACE ATTACH "EXPLOSIVES A" PLACARDS TO BOTH SIDES FRONT AND BACK OF THE TRAILER AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER

15 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

FTL LTL (35, 40, 42, 44, 45 AND 48 FT TRAILERS)

1 A FULL TRUCKLOADING CONSISTS OF FOUR VLS CANISTERS AND WEIGHS 29,337 POUNDS

2 THE PREPOSITIONED BLOCKING SHOWN IN DETAIL A MUST BE REINSTALLED ON THE TRAILER FLOOR BEFORE LOADING THE CANISTERS

NOTE

THE SKID ENDS OF THE CANISTER MUST BE IN THE "UP" POSITION
FOR CANISTERS TO FIT OVER THE PREPOSITIONED BLOCKING

3 THREE VLS CANISTERS COMPRISE AN UNSTABLE LOAD AND CANNOT BE LOADED

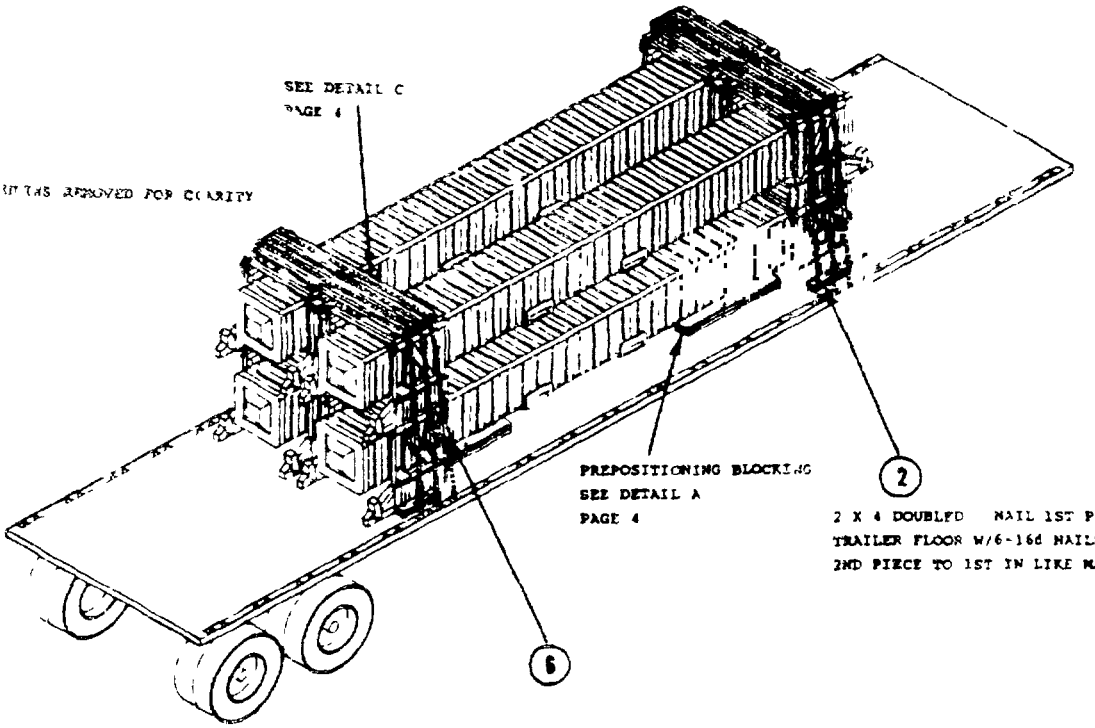
4 EMPTY OR EXPENDED MK 14 CANISTERS SHALL ALSO BE SHIPPED IN ACCORDANCE WITH THIS STD

NOTE

TARPAULINS MUST BE USED FOR ALL TRUCKLOADS DEPICTED IN THIS
MIL-STD

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NOTE TARP NOT SHOWN FOR CLARITY



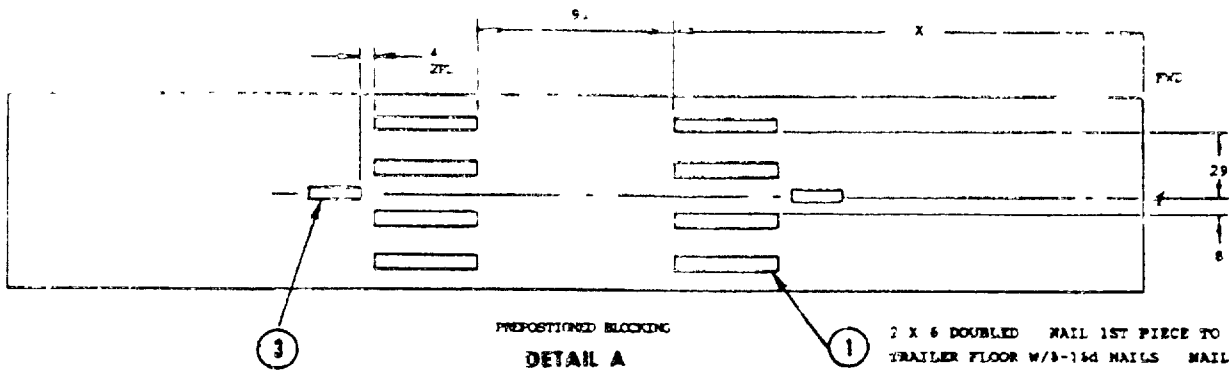
2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

NOTE ALL OVER THE ROAD SHIPMENTS OF THIS MISSILE MUST INCLUDE THE SECURITY SAFEGUARDS AND PHYSICAL PROTECTION REQUIRED BY JCMP INSTRUCTION 4601 1A

6	6	CHAIN AND LOAD BINDER		5/16 OR 3/8
4	5	SEPARATOR CLEAT	DETAIL C	2 X 4 X 36
12	4	PROTECTOR BOARD	DETAIL C	2 X 6 X 84
4	3	SPACER CLEAT		2 X 4 X 24
8	2	SLEEPER		2 X 4 X 24
16	1	BACKUP CLEAT		2 X 6 X 8
REQ'D	ITEM	DESCRIPTION	MATERIAL	DIMENSIONS

LIST OF MATERIALS

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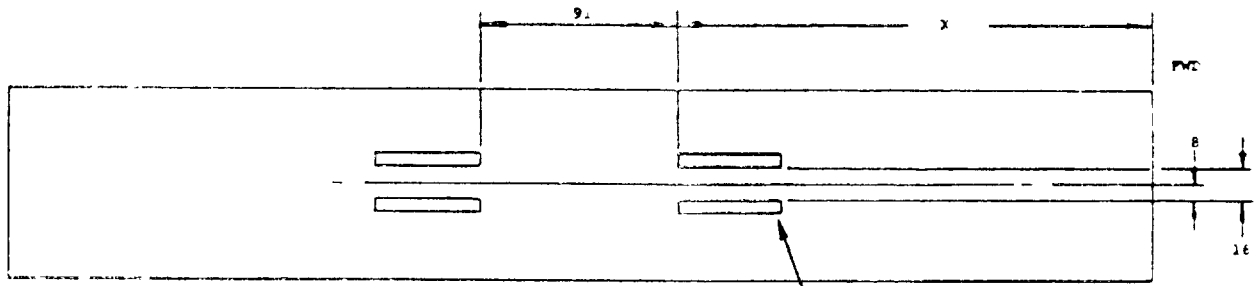


PREPOSITIONED BLOCKING

DETAIL A

(3) 2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

(1) 2 X 6 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/8-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

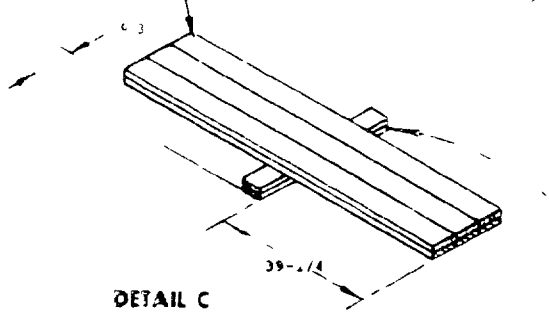


PREPOSITIONED BLOCKING

DETAIL B

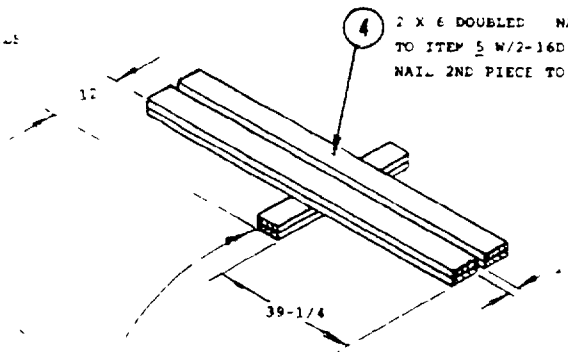
(1) 2 X 6 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/8-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

(4) 2 X 6 DOUBLED NAIL 1ST PIECE TO ITEM 5 W/2-16D NAILS/JOINT NAIL 2ND PIECE TO 1ST W/6-10D NAILS



DETAIL C

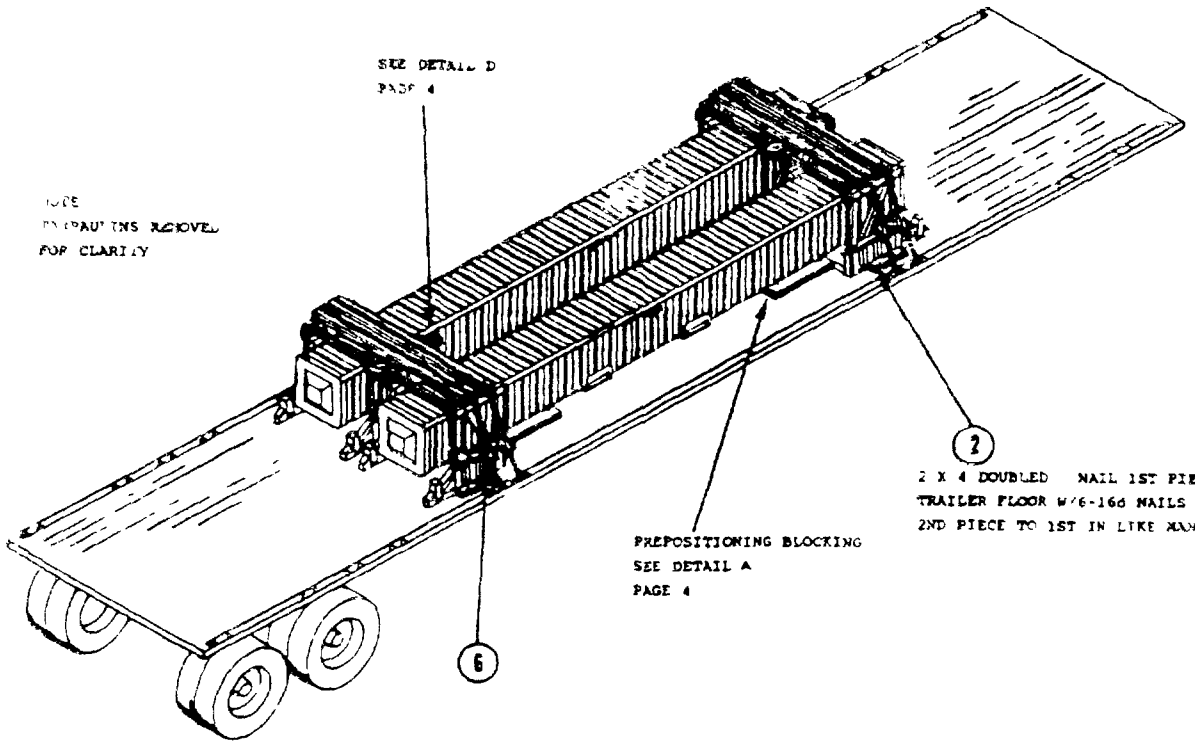
(4) 2 X 6 DOUBLED NAIL 1ST PIECE TO ITEM 5 W/2-16D NAILS/JOINT NAIL 2ND PIECE TO 1ST W/6-10D NAILS



DETAIL D

(5)

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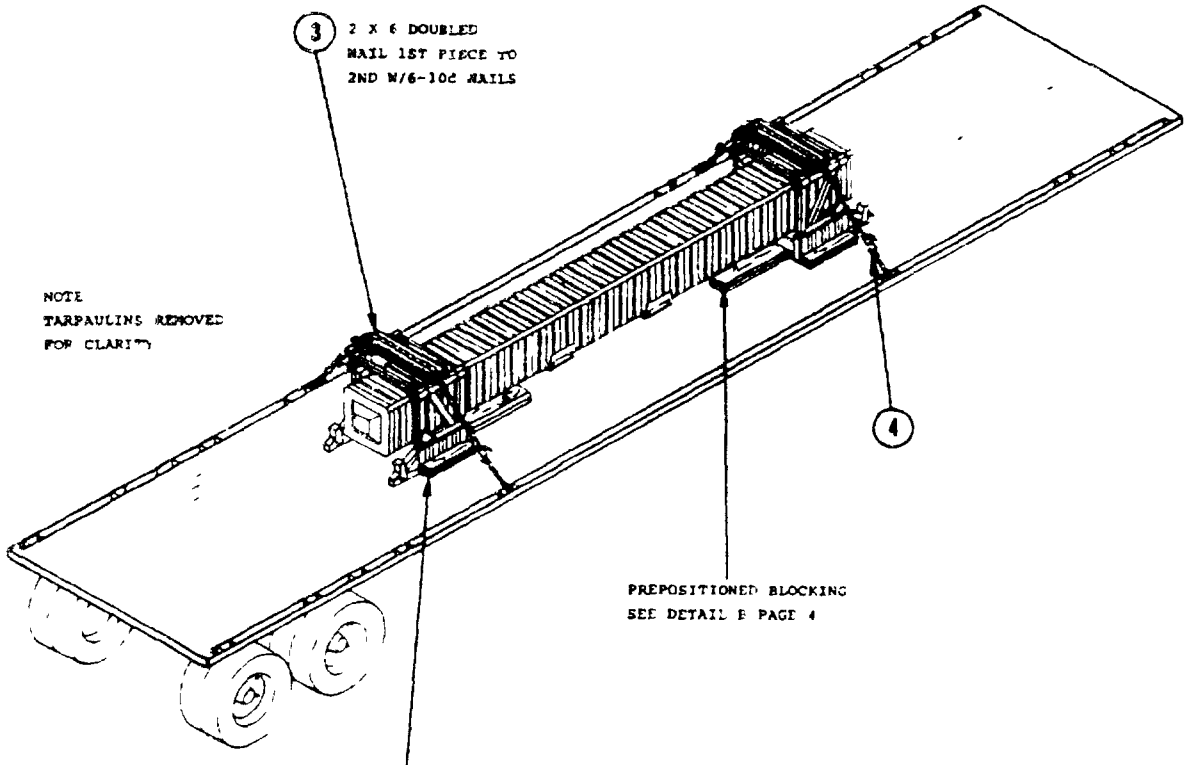
TRUCKLOAD DATA

NUMBER OF CONTAINERS 2
WEIGHT 14 6[illegible] LB

REQ'D	ITEM	DESCRIPTION	MAT'L/SPEC	REMARKS
2	6	CHAIN AND LOAD BINDER		5/16 OR 3/8
4	5	SEPARATOR CLEAT	DETAIL D	2 X 4 X 36
8	4	PROTECTOR BOARD	DETAIL D	2 X 6 X 84
4	3	SPACER CLEAT		2 X 4 X 24
8	2	SLEEPER		2 X 4 X 24
16	1	BACKUP CLEAT		2 X [illegible] X 36

LIST OF MATERIALS

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TRUCKLOAD DATA

NUMBER OF CONTAINERS -
WEIGHT 7 333 LB

QTY	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
2	4	CHAIN AND LOAD BINDER		5/16 OR 3/8
4	3	PROTECTOR BOARD		2 X 6 X 42
8	2	SLEEPER		2 X 4 X 24
8	1	BACKUP CLEAT		2 X 6 X 48

LIST OF MATERIALS

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO 8140 (SH845)

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